



Prostate cancer detection using Henry firefly gas solubility optimization-based deep residual network

Siva Kumar Reddy¹ · Kalaivani Kathirvelu¹

Received: 22 May 2023 / Revised: 27 July 2023 / Accepted: 23 August 2023 /

Published online: 12 September 2023

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023

Abstract

Accurate and appropriate prostate cancer detection can significantly reduce the death rate. In this research, the Henry Firefly Gas Solubility Optimization (HFGSO)-based Deep Residual Network (DRN) is established for the autonomous detection of prostate cancer. The pre-processing is done by Cuckoo Search-Based (T2FCS) filter and Type 2 Fuzzy. Subsequently, segmentation is exhibited by devised multi-objective SegNet scheme. The multi-objective SegNet method is newly designed by updating the objective function of SegNet with loss function. The multi-objective SegNet is trained by HFGSO. Then, data augmentation is done with cropping and rotation, which improves the performance of detection. At last, cancer identification is executed with DRN, and it is trained by HFGSO. The developed optimized multi-objective SegNet with DRN technique also achieved increased performance for the detection of cancer, with a sensitivity, specificity, and accuracy 0.9367, 0.9130, and 0.9263.

Keywords Firefly algorithm · Deep residual network · Dice coefficient · Prostate cancer · SegNet

1 Introduction

The male reproductive part is the prostate, which is the gland and it presents under the bladder. Its major operation is to hide the fluid of the prostate. Generally, prostate cancer generates cells of the prostate gland. This cell affects the close by organs and tissues otherwise it widens to other organs of the body [15, 7]. In worldwide, this cancer is the fifth directing source of cancer death and the second most well-known cancer for men [1]. In addition, this cancer is the most commonly detected kind of cancer among men in more countries [16] it is reported that nearly 1 in 5 cancer detection [17]. In the globe, it is calculated that 359,000 allied deaths ensued in 2018 and 1.3 million new cases of prostate cancers [5]. The most accurate and

✉ Siva Kumar Reddy
sivakumarreddybojja@gmail.com

¹ Department of Computer Science and Engineering, Vels Institute of Science Technology and Advanced Studies (VISTAS), Chennai, Tamilnadu, India

sensible classification of prostate cancer can strongly minimize the cancer death rate. For estimating the early identification of prostate cancer, a high serum level of Prostate Specific Antigen (PSA) is considered as extensive biomarker for needle biopsy [10, 1].

Magnetic Resonance Imaging (MRI) is appeared as the substitute for clinical standard Transrectal Ultrasound (TRUS) for detecting cancer, staging, focal therapy, biopsy guidance and localization. In recent days, multi-parametric MRI (mp-MRI) is the most indispensable role in prostate cancer [30] evaluation. Besides, mp-MRI can offer diffusion, metabolic and perfusion information of prostate tissue along with soft tissue contrast, which enhances the detection of cancer areas [6]. Furthermore, mp-MRI has been included for enhances the sensitivity and possibly decrease the unwanted biopsies by a quarter and lesser the over detection, while compared with standard TRUS [18]. Prostate cancer evaluation with mp-MRI is labour exhaustive process and it needs a high level of experience. Even though, the inspecting period for mp-MRI is above thirty minutes [19]. Bi-parametric MRI (bp-MRI) has developed as a quicker alternative short of reducing the detection accuracy of prostate cancer [20]. The Diffusion Weighted Imaging (DWI) and Type 2 (T2-weighted scans are essential for bp-MRI, which decreases the scanning duration as 17 minutes [19]. The bp-MRI has similar detection of precision towards mp-MRI [23, 21, 5, 3].

In current days, several investigations develop deep learning methods in medical images for categorizing inequalities. The dissimilar features are extracted from physiological, biological and neuro-physiological systems formerly in order to study the underlying system [22]. Moreover, hybrid features were also extracted [32] using morphology, elliptical shapes, geometry, complex dynamics, noise robustness, and texture properties [29], colon, prostate and breast [28] cancers. In addition, the pipeline depends on Convolutional Neural Network (CNN) is developed in order to identify prostate cancer [10, 6]. The medical imaging research community has moved their interest towards deep learning-enabled techniques along with promising outcomes of CNN in the computer vision area to advance Computer-Aided Diagnosis (CAD) tools for detecting cancer [2]. Moreover, Residual Networks (ResNets) are capable of easily and quickly training the deep neural networks [8]. The better risk stratification, detection accuracy, focal therapy and active surveillance are some of the clinical key challenges for the prostate cancer detection process. These problems and difficulties with prevailing prostate cancer detection strategies considered as a motivation for finding an innovative approach.

The contribution of the research:

- **HFGSO-driven DRN for prostate cancer detection:** An effectual cancer detection model is introduced by means of optimized multi-objective SegNet with DRN. The devised optimized multi-objective SegNet is newly developed by varying the objective function of SegNet with pixel-wise cross entropy and dice coefficient. The optimization model trained the multi-objective SegNet which is termed as HFGSO model. The HFGSO method is formulated by integrating Firefly algorithm (FA) and Henry gas solubility optimization (HGSO). Finding of cancer is completed by HFGSO trained DRN.

2 Literature Survey

In order to identify prostate cancer Wang, Y., et al. [1] devised a method namely stacking driven ensemble learning model. In this model, the random forest was employed as a stacking approach for sub-model combinations. Moreover, a rule extraction process was introduced for extracting effectual detection rules from the trimmed ensemble technique. This

method effectively replaced the time-consuming cross-validation and obtained improved detection accuracy. However, this model consumed more training time than single classifiers. To reduce the training time, Yoo, S., et al. [2] obtained Deep CNN (Deep CNN) for prostate cancer identification. Here, Diffusion-Weighted MRI (DWI) was considered as input and it was pre-processed. The classification results were greatly maximized in this method, although this technique is not succeeded to include lesions sequentially in nearby slices for reducing processing time. For decreasing the processing time and for identifying prostate cancer by means of MRI, Yu, X., et al. [3] introduced a scheme called as Panoptic. In this algorithm attention module was applied for optimally integrating global and local image features. This approach has a better capability to combine corresponding information from dissimilar sources. Even though, this scheme was not appropriate for other medical imaging applications. In order to consider other medical imaging applications, Cao, R., et al. [4] modelled multi-class CNN, termed Focal-Net for prostate cancer detection. To eradicate sounds from the input image, the first MRI image was first taken and pre-processed. Next data augmentation was executed for an effective training process. The fine-grained Gleason score groups and prostate cancer are effectively identified by this technique. Lesion information in the imaging plane is hard to confirm.

To easily identify lesion segmentation, De Vente, C., et al. [5] developed Neural Network (NN) for prostate cancer detection. The ROI was mined from pre-processed imager for defective detection. This technique performed effectual grading and detection of prostate lesions, but still, this model has a limited number of databases. To increase the performance even in the huge dataset, Xu, H., et al. [6] presented a Residual network for detecting prostate cancer. The MRI image was taken as input and it was segmented for effectual detection process. The method highly decreases the time required by radiologists for understanding the image, although prostate regions were not accurately segmented. For precise segments of the prostate portions, Duran-Lopez, L., et al. [7] deep learning-driven CAD model was implemented for prostate cancer identification. Here, the novel patch-scoring technique was applied to performing the pre-processing process, which eliminates the noises from the input image. Meanwhile, patches were employed as input for CNN, wherein the cancer was detected. This approach easily highlights the malignant tissue, even though the data size was limited. To increase the data size, Lee, W., et al. [8] devised CNN for prostate cancer detection. In this method, a Raman spectrum from Extracellular Vesicles (EVs) was obtained for detecting cancer. This scheme had a high degree of classification accuracy; however computational complexity was not decreased.

3 Optimized Multi-objective SegNet with DRN for Cancer Segmentation and Detection

In this research, initially, the input image is obtained from a dataset, and the ROI [27] is extracted for further processing, then to reduce the noises and outliers present in the input image pre-processing is performed. The T2FCS filter [14] is employed for pre-processing, which effectively increases the quality of the input image. Meanwhile, segmentation is carried out using optimized multi-objective SegNet [25], which is designed with adjusting the objective function of SegNet with Pixel-wise cross entropy and dice coefficient [26]. Moreover, the newly developed multi-objective SegNet method is trained by HFGSO algorithm. The HFGSO model is newly developed by integrating FA [9, 11] with the HGSO model [12]. Finally, cancer identification is completed on the basis of DRN, and it is trained by means of the HFGSO. Figure 1 conveys the schematic view of the implemented approach.

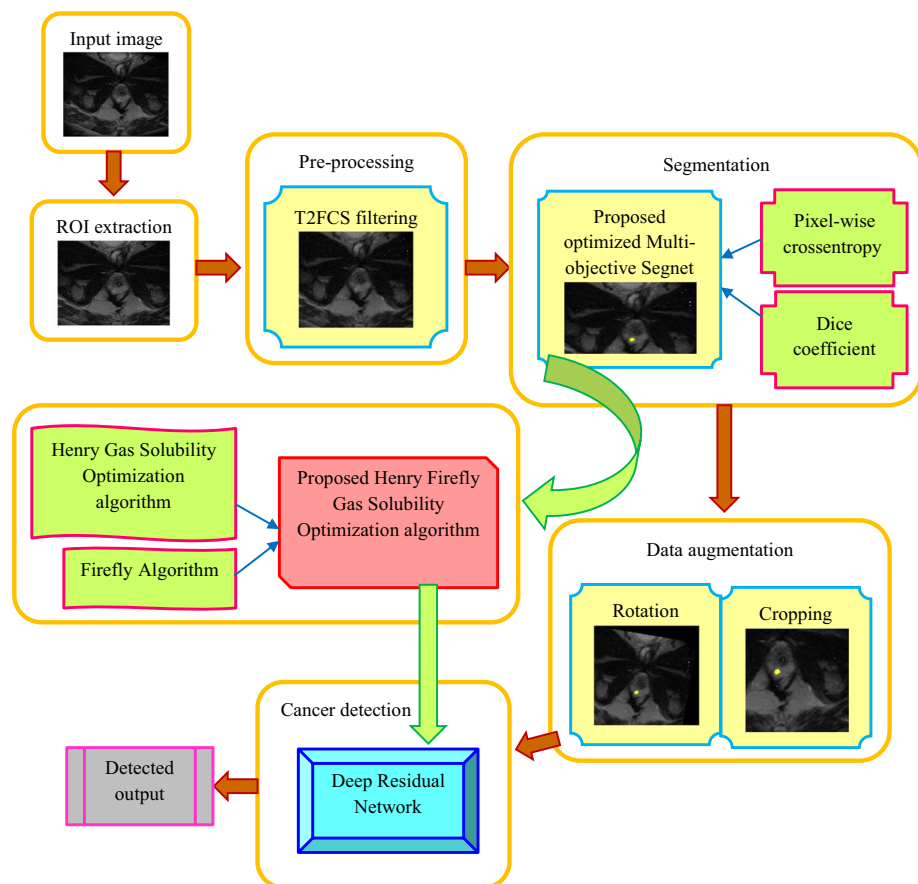


Fig. 1 Block diagram of segmentation and cancer detection model based on designed Optimized Multi-objective SegNet with DRN

3.1 Input Image

Let us imagine a dataset D with overall g images that is denoted as,

$$D = \{L_1, L_2, \dots, L_t, \dots, L_g\} \quad (1)$$

Where, L_t specifies input image at t^{th} index.

3.2 Extraction OF ROI

The input figure L_t is assumed as input for ROI extraction [27], which extracts the specific region of the image, thus computational difficulty is decreased. The ROI is described as the region of interest that is bound by pixel intensity rate. The outer portions of the input are eradicated in order to get rid of external deceptions. The ROI extracted image is conveyed as H_t .

3.3 Pre-processing

For pre-processing section, H_i is considered as an input, whereas the T2FCS filter is useful for eradicating the noises. The computational cost of this filter is highly decreased; thus, it is used for devised prostate cancer segmentation and detection. T2FCS approach [14] is designed by incorporating Cuckoo Search (CS) optimization model with type II fuzzy structure wherein each member is termed with respect to fuzzy as 1 and 0. F_i is the pre-processed image which is further used for segmentation.

3.4 Segmentation

After the pre-processing stage, using an optimized multi-objective SegNet model [25] the segmentation process is completed. Separating the images into several segments is a procedure for segmentation, which effectively decreases the complexity of the image. The segmentation is done using F_i .

3.4.1 Optimized multi-objective SegNet model

The optimized multi-objective SegNet [25] approach is established by considering objective functions, like cross entropy and dice coefficient with SegNet structure.

SegNet It is an encoder-decoder structure. Usually, encoder encompasses 13 convolutional layers in which the weights are trained. To obtain the upsampling in a non-linear manner the decoder executes pooling indices for the computation of the max-pooling of linked encoder. The output is denoted as N_i . The SegNet model is illustrated in Fig. 2.

3.4.2 Function of loss

Semantic segmentation is conserved as a pixel-level classification operation thus, it frequently implements pixel-wise cross entropy and dice coefficient [26] as loss function. These two functions are elucidated as follows,

i) *Pixel-wise cross-entropy*

The segmentation is considered as pixel-level identification task, thus it frequently implements pixel-wise dice coefficient and cross-entropy [26] as a function of loss. The pixel-wise cross-entropy is expressed as,

$$P(k, l) = \sum_s k_s \log l_s \quad (2)$$

ii) *Dice coefficient*

Moreover, the dice coefficient [26] is referred as estimating the performance of segmentation, and it is representing overlapping between two samples. The dice coefficient is given by,

$$H(k, l) = \frac{2 \sum_s l_s \times k_s}{\sum_s l_s + \sum_s k_s} \quad (3)$$

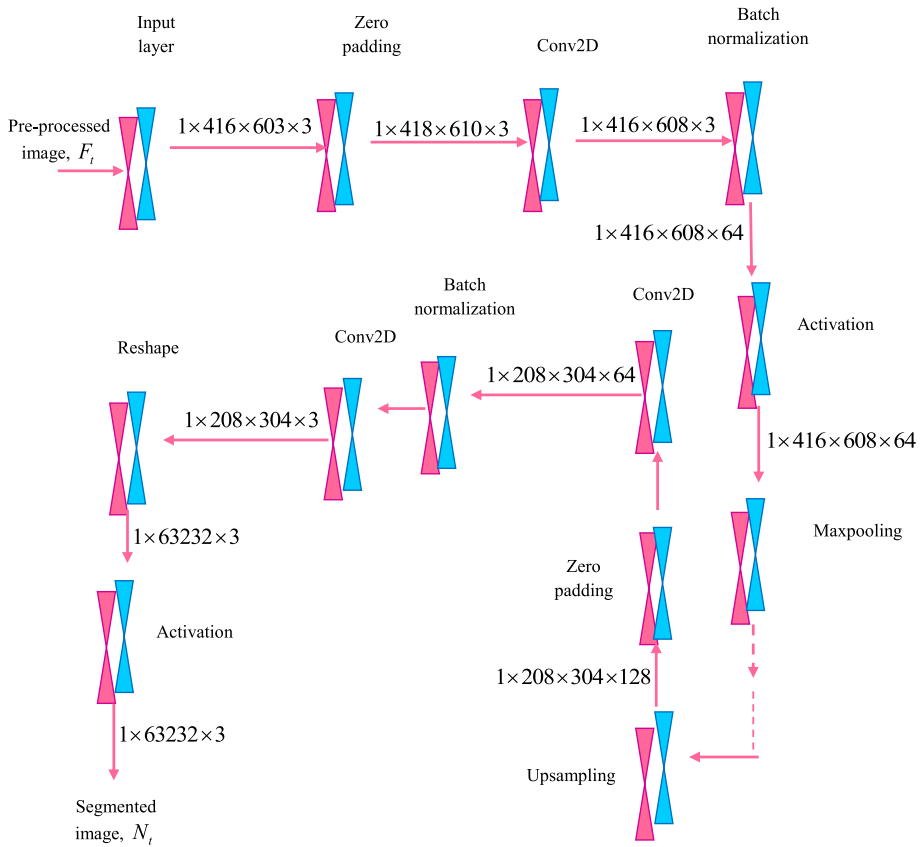


Fig. 2 Architecture of SegNet

Where, s represents category, k_s denotes label for prediction, and l_s indicates true label. Finally, the fusing is denoted as below,

$$M(k, l) = (1 - \mu) \left[\sum_s k_s \log(l_s) \right] - \mu \log \left(\frac{2 \sum_s l_s \times k_s + I}{\sum_s l_s + \sum_s k_s + I} \right) \quad (4)$$

Where, I implies smooth, which is a minimum number for instance $1e - 15$ and $\mu = 0.75$.

3.4.3 Proposed HFGSO for training process of SegNet

After that, through the optimization method, named HFGSO, the training of formulated multi-objective SegNet is done. The devised HFGSO algorithm is newly developed by integrating the HGSO method [12] and FA [9, 11]. The FA mainly includes two basic functions, including attracting mating partners and attracting potential grey. Here, flashing helps as a protective warning model. Furthermore, FA is depending on three idealized conditions, like each firefly is unisex, the accuracy of the firefly is exaggerated or recognized through a landscape of objective function, and their magnetism

is proportional to brightness. This model was not effective for solving multi-objective optimization function. Hence, the HGSO approach is integrated with FA, thereby optimization issues are solved effectively. The methods of the designed HFGSO technique are demonstrated as below,

1). Initialization

Originally, solution initialization is performed where the position of gases [12] and total gases is started, which is specified as,

$$R_h(p+1) = R_{\min} + d \times (R_{\max} - R_{\min}) \quad (5)$$

where, d implies random number among 0 and 1, R_h is the location of h^{th} gas in population y , $R_{\min}$ and $R_{\max}$ are bounds of problem, and p symbolizes iteration. The amount of gas h , the value of Henry's constant kind $g(E_h(p))$, the partial pressure $T_{h,g}$ of gas h , $\nabla_{\text{sol}} Y/Z$ constant value of $g(O_h)$ are initialized by below expression.

$$E_h(p) = n_1 \times \text{rand}(0, 1), T_{h,g} = n_2 \times \text{rand}(0, 1), O_g = n_3 \times \text{rand}(0, 1) \quad (6)$$

where, n_1, n_2, n_3 are constant values $n_1 = 5E - 02$, $n_2 = 100$, and $n_3 = 1E - 02$.

2). Clustering

In contrast to the number of gas groups, the population agents are grouped into equivalent groups. Additionally, every cluster contains equivalent gases, so it is associated with Henry's constant value E_h .

3). Fitness Estimation

The optimal solution is determined by the fitness function using Eq. (4).

4). Henry's coefficient Updation

Once estimating the function of fitness, the coefficient of Henry's is upgraded by,

$$E_h(p+1) = E_h(p) \times \exp(-O_g \times (1/W(p) - 1/W^\phi)); W(p) = \exp(-p/\nu) \quad (7)$$

Where, W refers to the temperature, Henry's coefficient is denoted as E_h , ν denotes the total amount of iterations, and W^ϕ signifies constant, which is equal to 298.15.

5). Solubility Update

The updating of solubility is carried out by,

$$A_{h,g}(p) = Z \times E_h(p+1) \times T_{h,g}(p) \quad (8)$$

Where, Z represents constant value, partial pressure on h in cluster g is represented as $T_{h,g}$ and $A_{h,g}$ denotes gaseous solubility of h in cluster g .

6). Upgrade the location

The location is upgraded by means of formulated HFGSO algorithm, which is illustrated as below,

$$R_{h,g}(p+1) = R_{h,g}(p) + N \times v \times \chi \times (R_{h,best}(p) - R_{h,g}(p)) + N \times v \times v \times (A_{h,g}(p) \times R_{best}(p) - R_{h,g}(p)) \quad (9)$$

$$R_{h,g}(p+1) = R_{h,g}(p) + N \times v \times \chi \times R_{h,best}(p) - N \times v \times \chi \times R_{h,g}(p) + N \times v \times v \times A_{h,g}(p) R_{best}(p) - N \times v \times v \times R_{h,g}(p) \quad (10)$$

$$R_{h,g}(p+1) = R_{h,g}(p) - N \times v \times \chi \times R_{h,g}(p) - N \times v \times v \times R_{h,g}(p) + N \times v \times \chi \times R_{h,best}(p) + N \times v \times v \times A_{h,g}(p) R_{best}(p) \quad (11)$$

$$R_{h,g}(p+1) = R_{h,g}(p)(1 - Nv\chi - Nv) + Nv(\chi R_{h,best}(p) + vA_{h,g}(p)R_{best}(p)) \quad (12)$$

Here, the movement of FA [11] is combined for solving optimization issues, thus,

$$R_{h,g}(p+1) = R_{h,g}(p) + \lambda_0 e^{-\psi s_{hg}^2} (R_g(p) - R_h(p) + \phi \mathfrak{R}_h) \quad (13)$$

$$R_{h,g}(p+1) = R_{h,g}(p) + \lambda_0 e^{-\psi s_{hg}^2} R_g(p) - \lambda_0 e^{-\psi s_{hg}^2} R_h(p) + \phi \mathfrak{R}_h \quad (14)$$

$$R_{h,g}(p) = \frac{R_{h,g}(p+1) - \lambda_0 e^{-\psi s_{hg}^2} R_g(p) - \phi \mathfrak{R}_h}{(1 - \lambda_0 e^{-\psi s_{hg}^2})} \quad (15)$$

Substitute Eq. (15) in (12),

$$R_{hg}(p+1) = \left(\frac{R_h(p+1) - \lambda_0 e^{-\psi s_{hg}^2} R_g(p) - \phi \mathfrak{R}_h}{(1 - \lambda_0 e^{-\psi s_{hg}^2})} \right) (1 - Nv\chi - Nv) + Nv(\chi R_{h,best}(p) + vA_{h,g}(p)R_{best}(p)) \quad (16)$$

$$R_{hg}(p+1) - \frac{R_h(p+1)}{(1 - \lambda_0 e^{-\psi s_{hg}^2})} (1 - Nv\chi - Nv) = \left(\frac{\lambda_0 e^{-\psi s_{hg}^2} R_g(p) R_h(p) + \phi \mathfrak{R}_h}{(1 - \lambda_0 e^{-\psi s_{hg}^2})} \right) (Nv\chi + Nv - 1) + Nv(\chi R_{h,best}(p) + vA_{h,g}(p)R_{best}(p)) \quad (17)$$

$$R_{hg}(p+1) - \frac{R_h(p+1)}{1 - \lambda_0 e^{-\psi s_{hg}^2}} \left(1 - \lambda_0 e^{-\psi s_{hg}^2} - 1 + Nv\chi + Nv \right) = \left(\lambda_0 e^{-\psi s_{hg}^2} R_g(p) + \phi \mathfrak{R}_h \right) (Nv\chi + Nv - 1) + Nv(\chi R_{h,best}(p) + vA_{h,g}(p)R_{best}(p)) \left(1 - \lambda_0 e^{-\psi s_{hg}^2} \right) \quad (18)$$

$$R_{hg}(p+1) = \frac{\left(\lambda_0 e^{-\psi s_{hg}^2} R_g(p) + \phi \mathfrak{R}_h \right) (Nv\chi + Nv - 1) + Nv(\chi R_{h,best}(p) + vA_{h,g}(p)R_{best}(p)) \left(1 - \lambda_0 e^{-\psi s_{hg}^2} \right)}{(Nv\chi + Nv - \lambda_0 e^{-\psi s_{hg}^2})} \quad (19)$$

where, χ implies scaling factor, v represents a random constant, $R_{best}(p)$ is the current best solution at p^{th} iteration, N denotes flag that varies the direction, $R_{h, best}(p)$ is best solution h in a cluster, λ_0 symbolizes attractiveness at $v=0$, $\mathfrak{R}_h$ specifies a random vector of integer being drawn from Gaussian distribution, and $A_{h, g}$ represents solubility of gas h in cluster g .

$$\chi = \varpi \exp\left(\frac{-N_{best}(p) + \tau}{N_{h, g}(p) + \tau}\right); \tau = 0.05 \quad (20)$$

Where, ϖ implies constant.

7). **Escape from local optimum**

This process is done to leave from local optimal, thus whole worst search agents are positioned and chosen by pursuing expression.

$$X_j = X \times (\text{rand}(z_2 - z_1) + z_1); z_1 = 0.1 \text{ and } z_2 = 0.2 \quad (21)$$

where, X conveys the search agents quantity.

8). **Upgrade the search agent location**

The search agents location is changed by the following eqn.

$$B_{h, g} = B_{minh, g} + \delta \times (B_{maxh, g} - B_{minh, g}) \quad (22)$$

Where, B_{min} and B_{max} symbolize bounds of problem, δ refers to random integer and $B_{h, g}$ denotes the gas location h in the cluster g .

9). **Evaluating feasibility of solution**

With Eq. (4) measure of fitness is calculated for each entity to achieve the function of fitness and optimal solution among the minimum value which is preferred as the optimal result.

10). **Termination**

For achieving the optimal solution, the beyond procedures are repeated constantly and thus the process gets completed after the estimation of maximum iterations. The pseudo-code of the HFGSO technique is specified in algorithm 1.

Algorithm 1 Pseudocode of HFGSO

Sl. No.	Pseudocode of HFGSO
1	Input: Amount of gas
2	Output: Optimal solution $R_h(p+1)$
3	Begin
4	Start location of gas and quantity
5	Several clusters forming by the population agents division
6	Classify cluster
7	Calculate the function of fitness
8	while $p < \text{maximum number of iteration}$ do
9	for every search agent do
10	Upgrade the location of all search agents by means of eq. (19)
11	end for
12	Upgrade gas types' Henry's coefficient by expression (7)
13	Change gas solubility based on eqn. (8)
14	Select and Rank the worst agents using eqn. (21)
15	Upgrade the location of worst agent with eqn. (22)
16	end while
17	$p = p + 1$
18	Return $R_h(p+1)$
19	End

Hence, the introduced HFGSO achieved better performance for segmenting pre-processed images in order to perform further processes. The segmented outcome is conveyed as C_t .

3.5 Data augmentation

The segmented image C_t is further used for the augmentation process, where cropping and rotation are applied. The data augmentation process is executed for maximizing the data size with no additional information.

3.5.1 Rotation

This model is applied to the segmented image, which rotates the image by -50 degree to 30 degree.

3.5.2 Cropping

This scheme is carried out in order for selecting a particular area, which crops the segmented image in one side by 30% .

Therefore, the data-augmented outcome is specified as V_r .

3.6 Cancer detection using Henry Firefly Gas Solubility Optimization-based DRN

The augmented data V_i is used for the cancer detection method. Here, DRN [13] is useful for detecting cancer for that augmented data is taken as input for DRN, which is trained by HFGSO.

3.6.1 Architecture of DRN

DRN technique [13] has a variety of layers such as linear classifier, convolutional layer, residual blocks and pooling. Usually, DRN attains learning performance and improved training under the constricted value of training data. It is operated for an efficient cancer recognition method. Data augmented output V_i is considered as input and architecture representation is shown in Fig. 3.

i). Convolutional layer

The 2D convolution layer is used in the training process to reduce unrestricted parameters. Here, the effective local receptive fields and weight-sharing performance are explained. The convolutional layer's input is controlled by filters known as kernels, and it applies computations to reduce the filter input matrix so that the kernel's dot product can be roughly estimated. The terms are shown as,

$$J2c(r) = \sum_{m=0}^{a-1} \sum_{w=0}^{a-1} B_{m,w} \cdot r(q+m)(s+w) \quad (23)$$

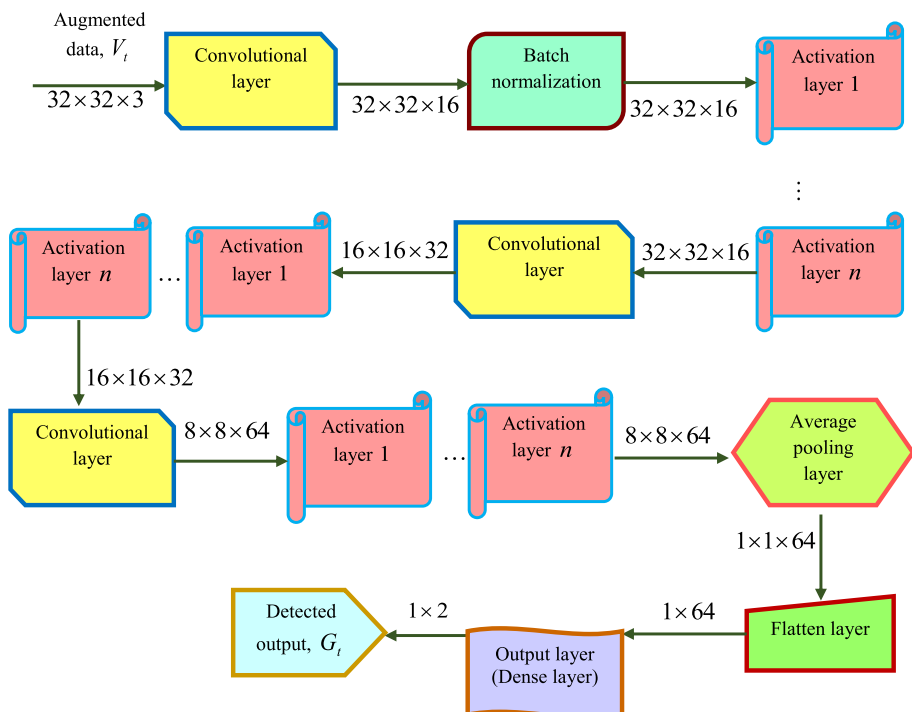


Fig. 3 DRN structural

$$J1c(r) = \sum_{y=0}^{J_m-1} B_y * r \quad (24)$$

Where, r conveys 2D outcome offered from the prior layer, q and s are useful in recording the coordinates in input, B_y is kernel size of y^{th} input neuron and m and s are area index in 2D kernel matrix, $*$ conveys cross-correlation operator and B denotes $a \times a$ kernel matrix.

ii) Pooling layer

It is correlated with the chronological convolutional layer to reduce the overfitting and spatial size of the feature map.

$$m_{out} = \frac{m_{in} - y_m}{z} + 1 \quad (25)$$

$$w_{out} = \frac{w_{in} - y_w}{z} + 1 \quad (26)$$

Where, m_{in} and w_{in} represent input 2D matrix width and height, y_w designates width for kernel size, y_m represents notes kernel size height and m_{out} and w_{out} specifies 2D matrix solution width and height.

iii) Activation function

It is the non-linear activation function to boost the features non-linearity and is utilized to develop the efficiency.

$$ReLU(r) = \begin{cases} 0, & r < 0 \\ r, & r \geq 0 \end{cases} \quad (27)$$

Where, r explicates input feature.

iv) Normalization OF Batch

The training set is divided into several tiny deposits, referred to as mini-batches, in this instance. These batches are employed for educational purposes. Switching between convergence and computational complexity is made easier by it. The activation factors are now modified and scaled to standardize input layers.

xxii) Residual Blocks

It designates an alternate relationship among the convolutional layers, whereas when compared with CNN this block includes the alternate route from input to output.

$$e = K(r) + r \quad (28)$$

Additionally, element matching is given as,

$$e = K(r) + P_r r \quad (29)$$

Where, P_r indicates dimension matching element, K is mapping relationship and r and e conveys the input and result of residual blocks.

vi) **Linear classifier**

It is the joining of Fully Connected (FC) layer and the softmax function. FC layer connects all neurons in every layer, and the normalizing is performed by softmax function. The DRN output is denoted as G_t .

$$e = P_{Y \times Z} r_{Z \times O} + U_{Y \times O} \quad (30)$$

where, $Y \times Z$ weight matrix is specified as $P_{Y \times Z}$, input feature is indicated as $r_{Z \times O}$ indicates, U conveys the bias.

3.6.2 Developed HFGSO for cancer detection

In addition, the training procedure is done for increasing detection performance by HFGSO. The introduced optimization algorithm, named the HFGSO is described already in Section 3.4.3. Here, $R_h \in P_{Y \times Z}$, U . The fitness measure for the cancer detection process is computed by Mean Square Error (MSE). The MSE with a minimum rate is believed as the best result.

$$\omega_f = \frac{1}{\epsilon} \sum_{t=1}^{\epsilon} (G_t^* - G_t)^2 \quad (31)$$

where, fitness function is indicated as ω_f , G_t is detected DRN outcome and G_t^* symbolizes target output. Therefore, the final cancer-detected output is represented as G_t .

4 Results and Discussion

The outcomes of devised HFGSO-driven DRN are explicated in this division.

4.1 Experimental Setup

With Intel i3 processor, Windows 10 OS, and 8GB RAM the cancer detection approach is utilized in the PYTHON tool. Here, the experimental parameters are selected using trial and error method.

4.2 Database Explanation

The designed prostate cancer detection approach is performed by means of a Prostate MRI dataset [24]. The Women's Hospital and Brigham, which offers multi-media presentations, provided the patient records from which the prostate MRI data was obtained. The database encompasses records of 230 patients with disease and detection information, which contains examination kind, exam explanation and examination date. The introduced method used the records of 20 patients.

4.3 Performance Metrics

The efficiency of the designed scheme is estimated based on specificity, sensitivity and accuracy.

4.4 Experimental Outcomes

The investigational conclusions of the introduced prostate cancer identification scheme are explicated in this division. The sample images for ROI extracted images are specified in Fig. 4b and the input images are given in Fig. 4a. The pre-processed images are

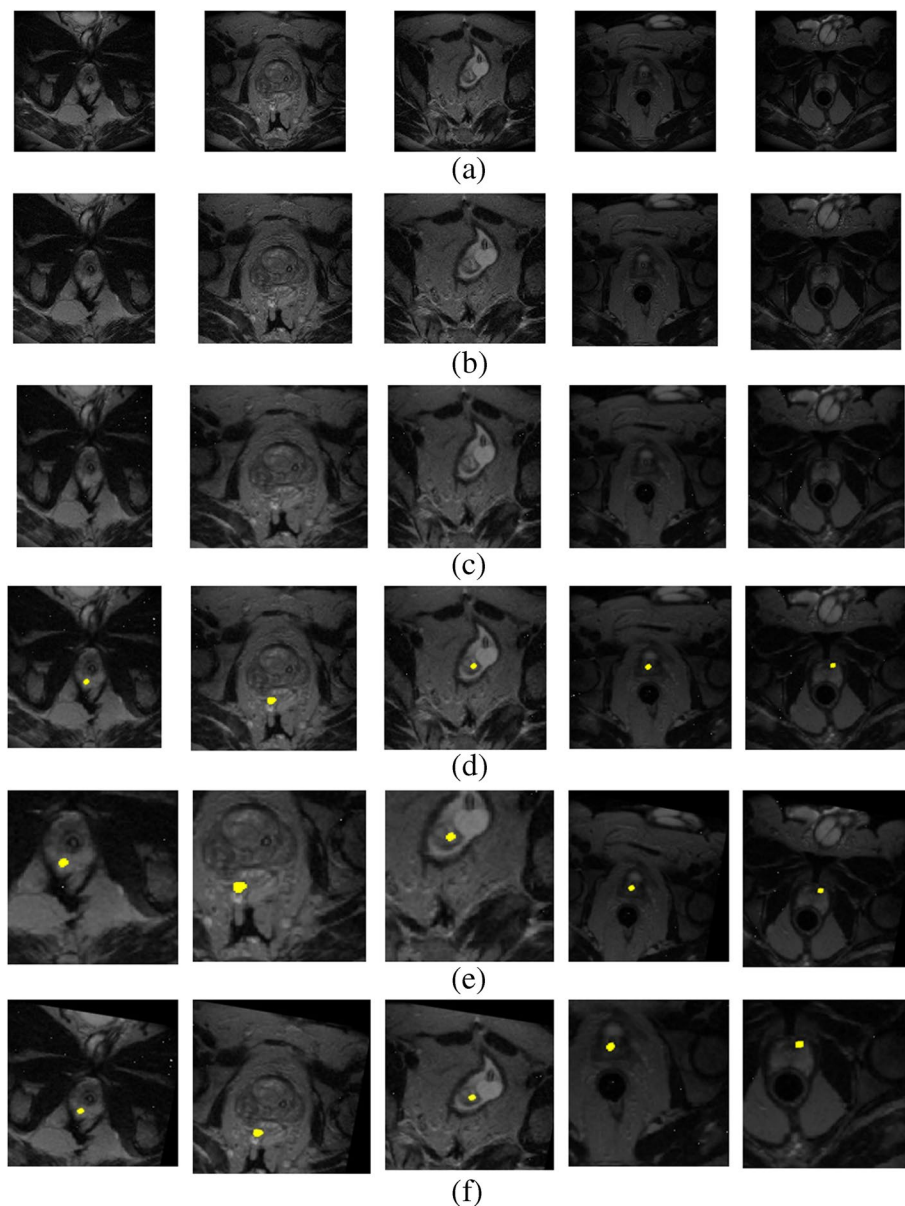


Fig. 4 Experimental outcomes (a) Input images (b) ROI extracted images (c) Pre-processed images (d) Segmented images (e) Rotated images (f) Cropped images

illuminated in Fig. 4c. Figure 4d denotes the segmented images, Figs. 4e and 4f convey the rotated and cropped images.

4.5 Performance Analysis

The analysis of HFGSO enabled DRN with performance metrics is shown in these divisions.

4.5.1 Analysis with training data percentage

Figure 5 specifies the performance study of HFGSO-driven DRN with training data percentage. The highest performance of the HFGSO-enabled DRN is obtained at 90% of training data and this is achieved by the effective integration of FA and HGSO. The best results are explained here. In Fig. 5a the efficiency investigation of accuracy is demonstrated. The accuracy of HFGSO-enabled DRN is 0.9238, 0.9249, 0.9253, and 0.9263 with iterations 5, 10, 15 and 20. Figure 5b signifies performance examination of the

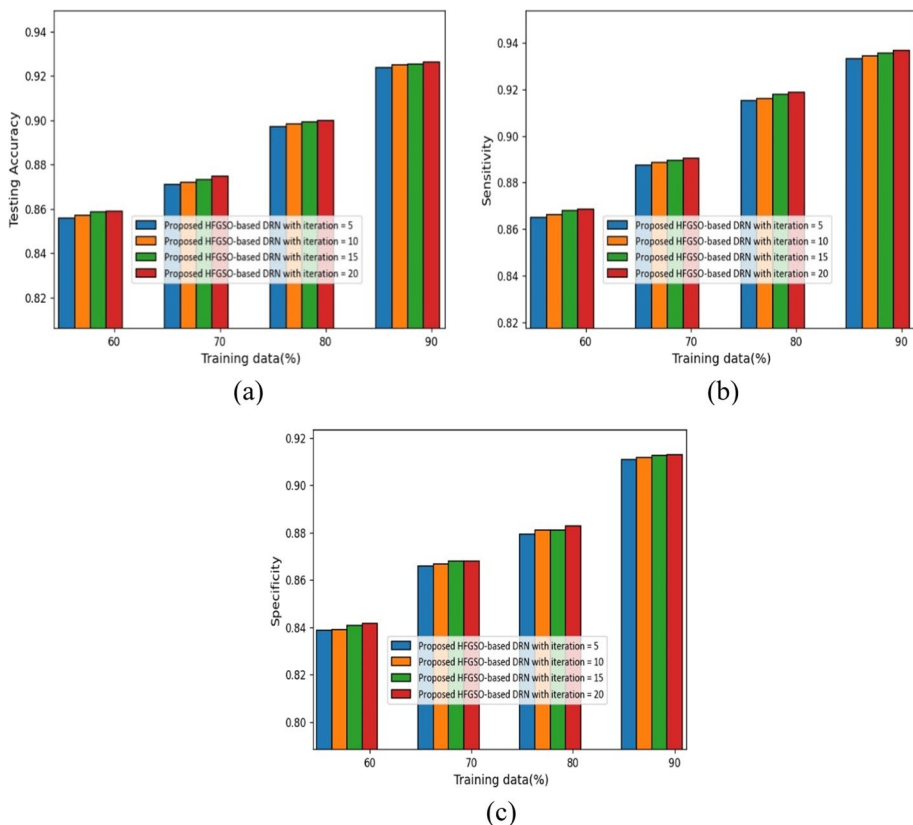


Fig. 5 Performance analysis of devised HFGSO-based DRN (a) Testing accuracy, (b) Sensitivity, (c) Specificity

designed approach with sensitivity is demonstrated. The sensitivity of HFGSO-based DRN is 0.9331, 0.9342, 0.9354 and 0.9367 with iteration 5, 10, 15, 20. The performance investigation of HFGSO-enabled DRN for specificity is represented in Fig. 5c. The specificity of HFGSO-enabled DRN is 0.9108, 0.9117, 0.9125, and 0.9130 with iterations 5, 10, 15 and 20.

4.5.2 Analysis with K-fold

The performance examination of the HFGSO-enabled DRN with k-fold value is shown in Fig. 6. The highest performance is attained at K-Fold=8 and iteration=20. The highest results are discussed here. Performance study of the designed approach with sensitivity is signified in Fig. 6a. The accuracy of HFGSO-enabled DRN is 0.9162, 0.9178, 0.9181 and 0.9192 with iterations 5, 10, 15 and 20. The performance analysis of HFGSO-enabled DRN for sensitivity is illustrated in Fig. 6b. The sensitivity of HFGSO-enabled DRN with iterations 5, 10, 15 and 20 is 0.9291, 0.9304, 0.9310, and 0.9326. In Fig. 6c the

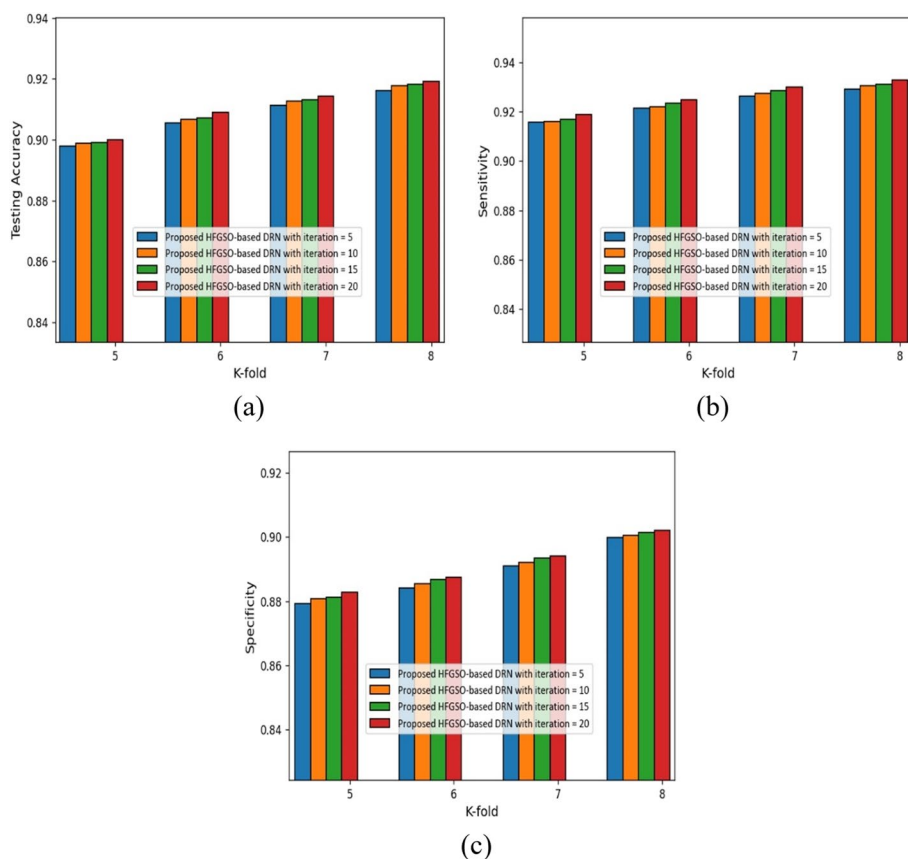


Fig. 6 Performance analysis of introduced HFGSO-driven DRN (a) Testing accuracy, (b) Sensitivity, (c) Specificity

performance examination of HFGSO-driven DRN with specificity is displayed. The specificity of HFGSO-driven DRN with iterations 5, 10, 15 and 20 is 0.8998, 0.9004, 0.9014, and 0.9021.

4.6 Comparative Analysis

The obtainable prostate cancer segmentation and detection, including DCNN [2], Panoptic model [3], Focal-Net [4], Residual NN [6], Deep learning driven CAD model [7], Snake Optimizer (SO)-based DRN [31], FA-based DRN, and HGSO-based DRN are taken into consideration for evaluating the performance of HFGSO-enabled DRN.

4.6.1 Analysis with training data percentage

Training data based comparative evaluation is given in Fig. 7. Figure 7a explicates the accuracy analysis. In 80% of data, the accuracy of HFGSO-driven DRN is 0.9000, and the efficiency development of HFGSO-driven DRN with DCNN, Panoptic technique, Focal-Net, Residual NN, Deep learning driven CAD model, SO-based DRN, FA-based DRN, and HGSO-based DRN is 16.95%, 14.15%, 6.81%, 4.15%, 3.93%, 3.67%, 3.42%, and 2.63%, respectively. Figure 7b denotes the comparative study of sensitivity. The sensitivity of

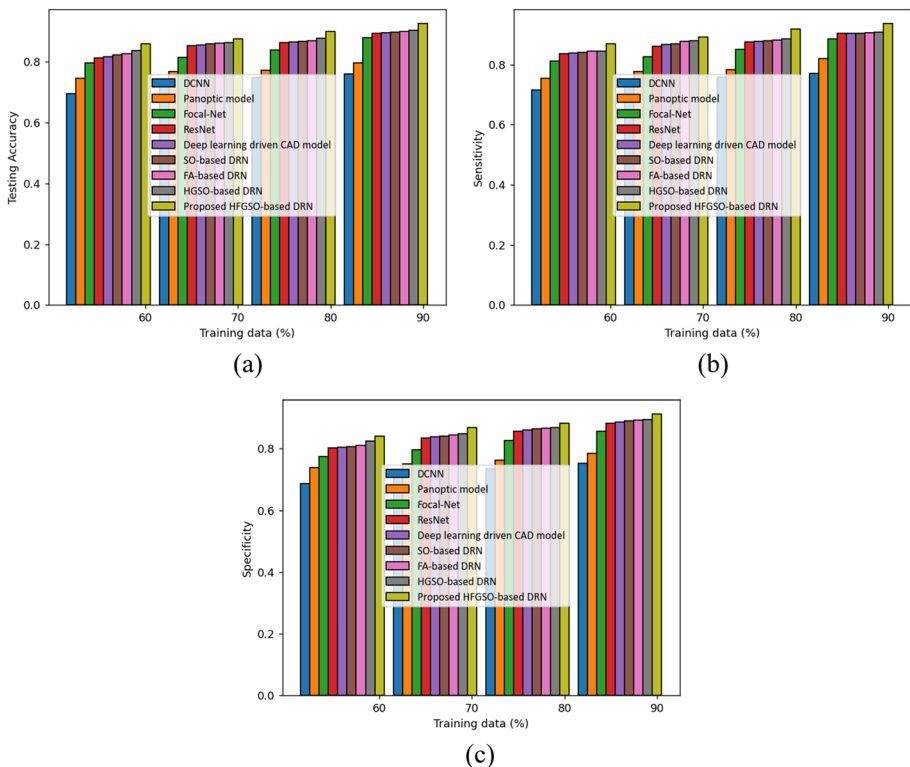


Fig. 7 Comparative analysis using training data (a) Testing accuracy, (b) Sensitivity, (c) Specificity

DCNN=0.7574, Panoptic technique=0.7828, Focal-Net=0.8506, Residual NN =0.8745, Deep learning driven CAD model is 0.8771, SO-based DRN is 0.8791, FA-based DRN is 0.8820, and HGSO-based DRN is 0.8841 and HFGSO on the basis of DRN is =0.91880 for the 90% of training data. The comparative study of specificity is demonstrated in Fig. 7c. Specificity of HFGSO-based DRN is 0.8828 for training data 80% and the efficiency development of HFGSO-driven DRN by means of DCNN, Panoptic technique, Focal-Net, Residual NN, Deep learning driven CAD model, SO-based DRN, FA-based DRN, and HGSO-based DRN is 16.75%, 13.51%, 6.44%, 2.92%, 2.58%, 2.12%, 1.74%, and 1.68%. The maximum performance of the model is achieved due to the effective DRN training with HFGSO.

4.6.2 Analysis with K-fold

K-Fold based comparative evaluation is given in Fig. 8. The accuracy is demonstrated in Fig. 8a. The accuracy of DCNN is 0.7573, panoptic technique is 0.7848, Focal-Net is 0.8497, Residual NN is 0.8750, Deep learning driven CAD model is 0.8784, SO-based DRN is 0.8792, FA-based DRN is 0.8810, and HGSO-based DRN is 0.8935, and HFGSO enabled DRN is 0.9142 for K-Fold=7. Figure 8b denotes the comparative study of sensitivity. When K-Fold=7 the sensitivity of HFGSO-based DRN is 0.9299 and the performance development of HFGSO-based DRN with DCNN is 17.58%, Panoptic approach

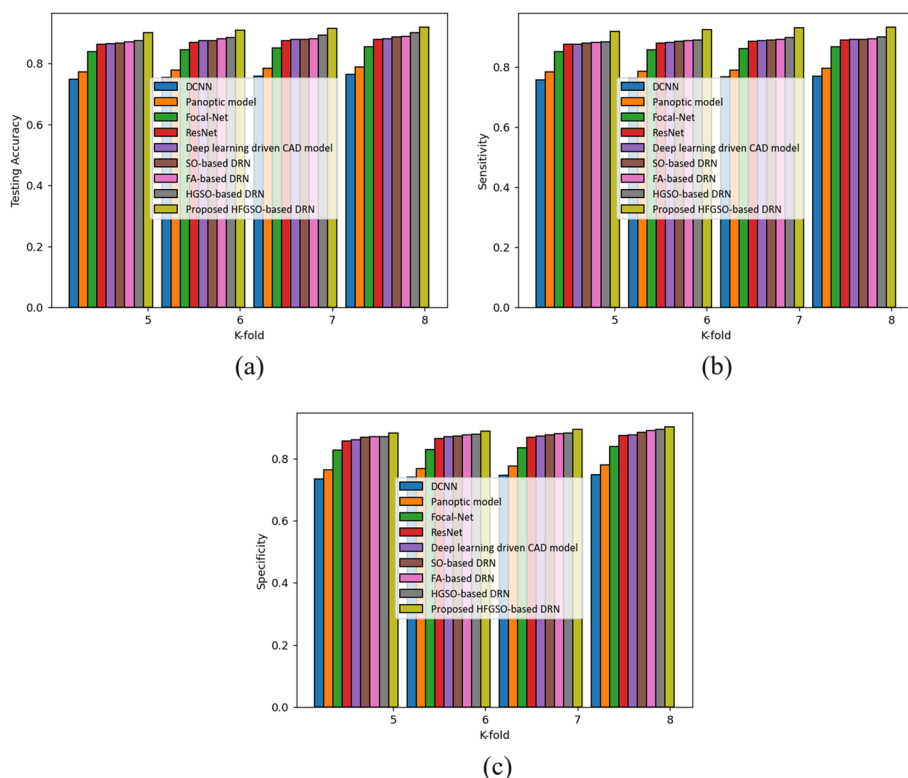


Fig. 8 Comparative analysis of K-Fold, (a) Testing accuracy, (b) Sensitivity, (c) Specificity

Table 1 Comparative discussion

Metrics	DCNN	Panoptic model	Focal-Net	ResNet	Deep learning driven CAD model	SO-based DRN	FA-based DRN	HGSO-based DRN	Proposed HFGSO-based DRN
Training data									
Accuracy (%)	76.05	79.70	87.92	89.39	89.48	89.71	89.90	90.43	92.63
Sensitivity (%)	77.02	81.96	88.51	90.29	90.31	90.41	90.51	90.71	93.67
Specificity (%)	75.30	78.42	85.73	88.35	88.71	89.01	89.20	89.54	91.30
K-Fold									
Accuracy (%)	76.42	78.80	85.50	87.90	88.14	88.65	88.93	90.01	91.92
Sensitivity (%)	77.00	79.57	86.69	88.90	89.10	89.24	89.41	90.01	93.26
Specificity (%)	74.85	78.00	83.94	87.30	87.67	88.45	88.93	89.35	90.21

The best results are indicated in bold font

is 15.01%, Focal-Net is 7.37%, Residual NN is 4.84%, Deep learning driven CAD model is 4.70%, SO-based DRN is 4.38%, FA-based DRN is 4.19%, and HGSO-based DRN is 3.58%. Figure 8c elucidates the comparative analysis of specificity. In K-Fold=7, the specificity of HFGSO-based DRN is 0.8940, while DCNN is 0.7457, Panoptic technique is 0.7758, Focal-Net is 0.8345 Residual NN is 0.8683, Deep learning driven CAD model is 0.8715, SO-based DRN is 0.8756, FA-based DRN is 0.8793, and HGSO-based DRN is 0.8820. In this research, the DRN is training by HFGSO and the HGSO approach is integrated with FA, thereby optimization issues are solved effectively. Thus, the efficiency of the HFGSO-based DRN is higher than the DCNN, Panoptic technique, Focal-Net, and Residual NN.

4.7 Comparative Discussion

The best results of all approaches are discussed in Table 1. The accuracy of prostate cancer detection replica is extremely improved through the effective data augmentation. Also, the multi-objective SegNet approach effectively increases the specificity of introduced prostate cancer detection. Additionally, by taking into account the HFGSO for the training of DRN, the specificity of the prostate cancer finding procedure is improved.

5 Conclusion

This manuscript gives an effective prostate cancer detection process using HFGSO on the basis of DRN. The extraction of ROI is fulfilled for input MRI for extracting significant regions. To eradicate the sounds from input MRI illustration the extracted ROI image is pre-processed. Segmentation is processed for pre-processed images in order to segment specific cancer regions for reducing the detection complexity. In this technique, multi-objective SegNet model is applied for segmenting essential regions. Moreover, cancer detection is done using DRN from segmented images. Additionally, the employed DRN and multi-objective SegNet model is trained by the HFGSO model. The HFGSO-based DRN attained the highest specificity of 0.9130, sensitivity of 0.9367 and testing accuracy of 0.9263. The HFGSO-based DRN is more accurate, so that it minimizes the cancer death rate. However, this method is not identified the kinds of prostate cancers, like Squamous cell carcinoma, Adenocarcinoma, Transitional cell carcinoma, and Small cell prostate cancer and this will be considered in the future work.

Acknowledgements I would like to express my very great appreciation to the co-authors of this manuscript for their valuable and constructive suggestions during the planning and development of this research work.

Author Contribution All authors have made substantial contributions to conception and design, revising the manuscript, and the final approval of the version to be published. Also, all authors agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Funding This research did not receive any specific funding.

Data Availability The data underlying this article are available in Prostate MRI Dataset, “<https://prostatemr.imagedatabase.com/>”.

Declarations

Conflict of Interest The authors declare no conflict of interest.

Informed consent Not Applicable.

Ethical approval Not Applicable.

References

- Wang Y, Wang D, Geng N, Wang Y, Yin Y, Jin Y (2019) Stacking-based ensemble learning of decision trees for interpretable prostate cancer detection. *Appl Soft Comput* 77:188–204
- Yoo S, Gujrathi I, Haider MA, Khalvati F (2019) Prostate cancer detection using deep convolutional neural networks. *Sci Rep* 9(1):1–10
- Yu X, Lou B, Zhang D, Winkel D, Arrahmane N, Diallo M, Meng T, von Busch H, Grimm R, Kiefer B, Comaniciu D (2020) Deep attentive panoptic model for prostate cancer detection using biparametric MRI Scans, In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, Cham, pp 594–604
- Cao R, Bajgirani AM, Mirak SA, Shakeri S, Zhong X, Enzmann D, Raman S, Sung K (2019) Joint prostate cancer detection and gleason score prediction in mp-MRI via FocalNet. *IEEE Trans Med Imaging* 38(11):2496–2506
- De Vente C, Vos P, Hosseinzadeh M, Pluim J, Veta M (2020) Deep learning regression for prostate cancer detection and grading in bi-parametric MRI. *IEEE Trans Biomed Eng* 68(2):374–383
- Xu H, Baxter JS, Akin O, Cantor-Rivera D (2019) Prostate cancer detection using residual networks. *Int J Comput Assist Radiol Surg* 14(10):1647–1650
- Duran-Lopez L, Dominguez-Morales JP, Conde-Martin AF, Vicente-Diaz S, Linares-Barranco A (2020) PROMETEO: A CNN-Based Computer-Aided Diagnosis System for WSI Prostate Cancer Detection. *IEEE Access* 8:128613–128628
- Lee W, Lenferink AT, Otto C, Offerhaus HL (2020) Classifying Raman spectra of extracellular vesicles based on convolutional neural networks for prostate cancer detection. *J Raman Spectrosc* 51(2):293–300
- Arora S, Singh S (2013) The firefly optimization algorithm: convergence analysis and parameter selection. *Int J Comput Appl* 69(3):48–52
- Iqbal S, Siddiqui GF, Rehman A, Hussain L, Saba T, Tariq U, Abbasi AA (2021) Prostate Cancer Detection Using Deep Learning and Traditional Techniques. *IEEE Access* 9:27085–27100
- Yang XS (2009) Firefly algorithms for multimodal optimization. In: *International symposium on stochastic algorithms*. Springer, Berlin, Heidelberg, pp 169–178
- Hashim FA, Houssein EH, Mabrouk MS, Al-Atabany W, Mirjalili S (2019) Henry gas solubility optimization: A novel physics-based algorithm. *Futur Gener Comput Syst* 101:646–667
- Chen Z, Chen Y, Wu L, Cheng S, Lin P (2019) Deep residual network based fault detection and diagnosis of photovoltaic arrays using current-voltage curves and ambient conditions. *Energy Convers Manag* 198:111793
- Kumar SV, Nagaraju C (2019) T2FCS filter: Type 2 fuzzy and cuckoo search-based filter design for image restoration. *J Vis Commun Image Represent* 58:619–641
- Garber K (2010) A tale of two cells: discovering the origin of prostate cancer. *J Natl Cancer Inst* 102(20):1528–1535
- Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A (2018) Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 68(6):394–424
- Siegel RL, Miller KD, Jemal A (2019) Cancer statistics. *CA Cancer J Clin* 69(1):7–34
- Ahmed HU, Bosaily AES, Brown LC, Gabe R, Kaplan R, Parmar MK, Collaco-Moraes Y, Ward K, Hindley RG, Freeman A, Kirkham AP (2017) Diagnostic accuracy of multi-parametric MRI and TRUS biopsy in prostate cancer (PROMIS): a paired validating confirmatory study. *Lancet* 388(10071):815–822
- Thestrup KCD, Logager V, Baslev I, Møller JM, Hansen RH, Thomsen HS (2016) Biparametric versus multiparametric MRI in the diagnosis of prostate cancer. *Acta Radiol Open* 5(8):2058460116663046
- Stanzione A, Imbriaco M, Coccoza S, Fusco F, Rusconi G, Nappi C, Mirone V, Mangiapia F, Brunetti A, Ragozzino A, Longo N (2016) Biparametric 3T Magnetic Resonance Imaging for prostatic cancer

- detection in a biopsy-naïve patient population: a further improvement of PI-RADS v2? *Eur J Radiol* 85(12):2269–2274
21. De Visschere P, Lumen N, Ost P, Decaestecker K, Pattyn E, Villeirs G (2017) Dynamic contrast-enhanced imaging has limited added value over T2-weighted imaging and diffusion-weighted imaging when using PI-RADSv2 for diagnosis of clinically significant prostate cancer in patients with elevated PSA. *Clin Radiol* 72(1):23–32
 22. Greff K, Srivastava RK, Koutník J, Steunebrink BR, Schmidhuber J (2016) LSTM: A search space odyssey. *IEEE Trans Neural Netw Learn Syst* 28(10):2222–2232
 23. Kuhl CK, Bruhn R, Krämer N, Nebelung S, Heidenreich A, Schrading S (2017) Abbreviated biparametric prostate MR imaging in men with elevated prostate-specific antigen. *Radiology* 285(2):493–505
 24. Prostate MRI Dataset "<https://prostatemrimagesdatabase.com/>", accessed on January 2022
 25. Badrinarayanan V, Kendall A, Cipolla R (2017) Segnet: A deep convolutional encoder-decoder architecture for image segmentation. *IEEE Trans Pattern Anal Mach Intell* 39(12):2481–2495
 26. Zhang H, Hong X, Zhou S, Wang Q (2019) Infrared image segmentation for photovoltaic panels based on Res-UNet. In: proceedings of Chinese Conference on Pattern Recognition and Computer Vision (PRCV), pp 611–622
 27. Renkalatha S, Suresh KV (2017) Automatic roi extraction in noisy medical images. *ICTACT J Image Vid Process* 7(4):1505–1514
 28. Fernandis JR (2021) ALOA: Ant Lion Optimization Algorithm-based Deep Learning for Breast Cancer Classification. *Multimed Res* 4(1):105–116
 29. Gokulkumari G (2020) Classification of Brain tumor using Manta Ray Foraging Optimization-based DeepCNN classifier. *Multimed Res* 3(4):32–42
 30. Mohammed Ismail B, Alam M, Tahernezahdi M, Vege HK, Rajesh P (2020) A machine learning classification technique for predicting prostate cancer. In: Proceedings of the 2020 IEEE International Conference on Electro Information Technology (EIT), IEEE, Chicago, IL, USA
 31. Hashim FA, Hussien AG (2022) Snake Optimizer: A novel meta-heuristic optimization algorithm. *Knowl-Based Syst* 242:108320
 32. Singh D, Kaur M (2022) Evolving fusion-based visibility restoration model for hazy remote sensing images using dynamic differential evolution. *IEEE Trans Geosci Remote Sens* 60(1):1–14

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.